

Gas Pressure, the Particle Model and Diffusion — Paper C

HARD • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every numerical answer must carry a unit. An answer with no unit does not score the final mark.
- Explanations must be written in terms of particles. “The particles get bigger”, “the particles expand” and “the particles push harder because they are squashed” score nothing.
- The mark for each question is shown in square brackets on the right.
- No formula is given on this paper, and no value for atmospheric pressure.
- Several questions describe apparatus you may not have seen before — apply the ideas you already have.
- Where an explanation is worth three marks or more, the mark scheme is looking for collisions per unit area per second, and, wherever the temperature changes, for the force of each collision as well.

SECTION 1 – TWO CHANGES, ONE AFTER THE OTHER

1. A sealed gas syringe holds 60 cm^3 of air at a pressure of 100 kPa and a temperature of 300 K.

- (a)** The plunger is pushed in until the volume is 20 cm^3 , at constant temperature. [3]
Calculate the new pressure and explain the result in terms of particles.

- (b)** The syringe is then locked at that volume and heated to 600 K. Calculate the pressure now. [1]

- (c)** State one assumption about the syringe that must be true for your answer to (b). [1]

- 2.** A sealed glass flask of gas is at a pressure of 100 kPa and a temperature of 300 K. It is cooled to 150 K. The flask does not change size.

(a) State and explain what happens to the pressure, in terms of particles. [2]

(b) Calculate the new pressure. [1]

(c) A student says the pressure falls “because cooling makes the particles smaller, so there is more space”. Explain what is wrong with this and give the correct reason. [2]

SECTION 2 – THE DIFFUSION TUBE

- 3.** A 60 cm glass tube has a cotton wool plug soaked in ammonia solution at the left-hand end and one soaked in concentrated hydrochloric acid at the right-hand end. Both are put in at the same moment. After several minutes a white ring of solid forms 36 cm from the ammonia end.

$$M_r(\text{NH}_3) = 17 \quad M_r(\text{HCl}) = 36.5$$

(a) Explain why a ring forms at all, and name the solid that forms it. [2]

(b) Calculate how far the hydrogen chloride travelled, and the ratio of the two distances. [2]

- (c)** Explain the ratio you calculated, in terms of the particles of the two gases. [2]

- 4.** The same experiment is repeated with the whole tube warmed from 20 °C to 60 °C.

- (a)** State and explain the effect on the time the ring takes to appear. [2]

- (b)** State and explain the effect on the position of the ring. [2]

SECTION 3 – THE LONG EXPLANATION

- 5.** A sealed metal can of air is left near a fire. Its volume cannot change. After a while the can bursts. [6]

Explain fully, in terms of particles, why the pressure inside rises as the can gets hotter, and why it eventually bursts. Write in continuous prose.

- 6.** A gas at a pressure of 250 000 Pa acts on one face of a piston of area 0.008 m². The atmosphere, at 100 000 Pa, acts on the other face.

(a) Calculate the force the trapped gas exerts on the piston. [1]

(b) Calculate the force the atmosphere exerts on the piston. [1]

(c) Calculate the resultant force on the piston and state its direction. [1]

(d) The piston is held still. State the size and direction of the force holding it. [1]

End of paper. Check every answer has a unit or a form the question actually asked for.